LESSON PLAN

ROCKET ENGINE (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	2	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to liquid rocket engines its advantages and disadvantages.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to liquid rocket engines its advantages and disadvantages.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain types of rocket engine and their working principle.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of liquid propellant storage diagrams; Safety: No handling of propellants; Supervision: Instructor monitors diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between solid and liquid propellants?
(b) Can you name one advantage of liquid propulsion over solid propulsion?
(c) What are the two main components of a liquid rocket engine?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of liquid rocket engine • Working principle of combustion chamber • Feed system types: Pressure and Pump • Propellant storage tanks • Thrust vector control mechanisms	L & Ds	PPT, IB	15 min
(b)	• Advantages: High specific impulse • Advantages: Throttleability and restart capability • Advantages: Precise thrust control • Disadvantages: Complex mechanical systems • Disadvantages: High maintenance requirements	L & Ds	PPT, PDF	20 min
(c)	• Comparison of LOX/Kerosene vs LOX/Liquid Hydrogen • Impact of propellant choice on engine performance • Safety considerations for liquid propellants • Future trends in liquid engine technology • Summary of key operational differences	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the pump-fed system differs from the pressure-fed system in a liquid rocket engine.
(b) List three distinct advantages and three distinct disadvantages of using liquid rocket engines compared to solid motors.
(c) Why is throttleability considered a critical advantage for launch vehicle guidance?

12. Home Assignment and Follow Up	Prepare a brief note comparing the specific impulse values of LOX/Kerosene and LOX/Liquid Hydrogen engines and explain how this affects mission design.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Instructor, P No.	Senior Instructor, Senior Instructor, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min